

Night Sky Star Classification using Neural Networks

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Business Problem

Astronomers and cosmologists face the challenge of distinguishing quasars, stars, and galaxies from vast amounts of telescope-collected celestial data. Quasars, though crucial for exploring the early universe, supermassive black holes, and cosmic structures, are rare and often misidentified due to overlapping features and imbalanced datasets. Manual classification is both labor-intensive and prone to mistakes. To address this, there is a growing need to automate and improve the accuracy of object classification through machine learning—especially neural networks. This approach aims to:

- Streamline astronomical research workflows
- Increase precision in quasar identification for cosmological analysis
- Minimize reliance on manual data labeling by harnessing intelligent, data-driven methods

Data Dictionary

Data Table

Data instances: 91000
Features: 10
Meta attributes: None
Target: Class 'class'

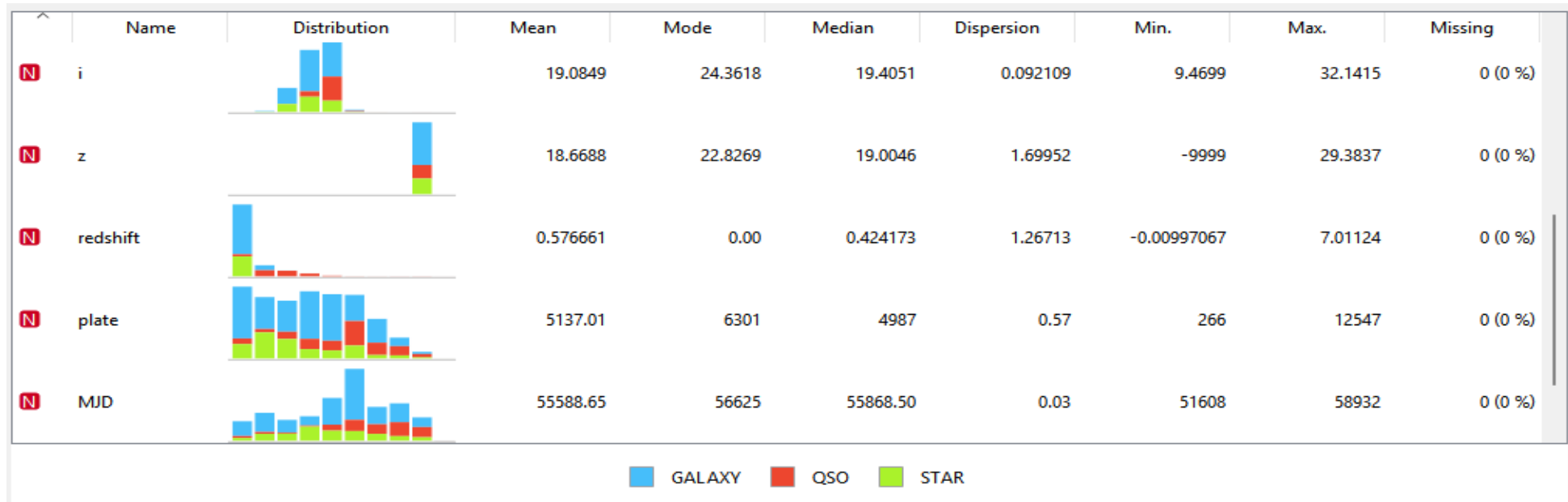
	class	alpha	delta	u	g	r	i	z	redshift	plate	MJD
1	STAR	123.679	17.7247	23.92281	23.44659	20.76838	20.1161	19.6687	4.76794e-05	2269	53711
2	QSO	6.61242	26.2872	20.61213	20.18499	20.05757	19.8047	19.7081	1.66542	7662	57358
3	GALAXY	217.895	51.4188	20.30099	18.47274	17.57094	17.1704	16.8381	0.0975175	1046	52460
4	GALAXY	161.69	24.0391	17.17042	15.37530	14.49252	14.0043	13.618	0.0449343	2481	54086
5	GALAXY	358.848	-10.5053	22.62808	20.13440	18.58930	17.9776	17.5416	0.289013	7166	56602
6	QSO	323.555	1.15041	22.48813	21.60654	21.21022	20.7127	20.4258	1.565	5144	55829
7	GALAXY	149.065	16.552	18.87258	17.79118	17.43896	17.2199	17.0514	0.0394904	2583	54095
8	QSO	158.735	39.6614	19.69144	18.24042	17.15510	16.8305	16.6928	0.150749	1432	53003
9	GALAXY	160.981	63.9649	23.74022	22.03101	20.31887	19.5347	19.1085	0.43418	7097	56667
10	GALAXY	242.072	41.9247	23.50154	21.73414	20.39247	19.4449	19.0666	0.482634	6044	56090
11	GALAXY	143.112	57.4818	22.12406	21.39937	21.85959	21.6674	21.6649	0.0294524	7445	56720
12	STAR	7.53755	-0.832536	21.37179	18.89540	17.42231	16.394	15.8265	-6.82421e-05	1493	52933
13	GALAXY	218.971	53.6871	21.00793	19.62881	18.17052	17.6581	17.3114	0.273978	6710	56416
14	STAR	228.326	20.6327	18.57260	17.44199	17.03977	16.891	16.8528	-0.000303112	2153	54212
15	GALAXY	125.066	51.0761	24.64049	22.17142	20.65574	19.6689	19.2187	0.547847	3695	55212
16	GALAXY	320.614	0.118614	22.71511	22.72988	22.26624	21.6557	22.3807	1.1054	9225	58013
17	QSO	226.115	54.6506	19.00792	18.85943	18.54663	18.4189	18.368	1.0647	793	52370
18	GALAXY	132.571	32.1027	22.38622	22.53691	21.41763	21.0776	20.1625	0.775895	10431	58137
19	GALAXY	330.635	12.6362	24.55610	22.02172	20.37426	19.5712	19.0575	0.432613	5062	55803
20	GALAXY	223.56	43.5783	24.40429	22.44450	20.62427	19.6123	19.0826	0.490584	6047	56098
21	STAR	258.531	30.646	25.35160	23.79435	22.43573	21.2246	19.2796	-0.00015323	11654	58543

Data Info

Data table properties

Name: star_classification
Size: ~100000 rows, 18 columns
Features: 17 numeric
Targets: categorical outcome with 3 classes
Missing data: none

Feature Statistics



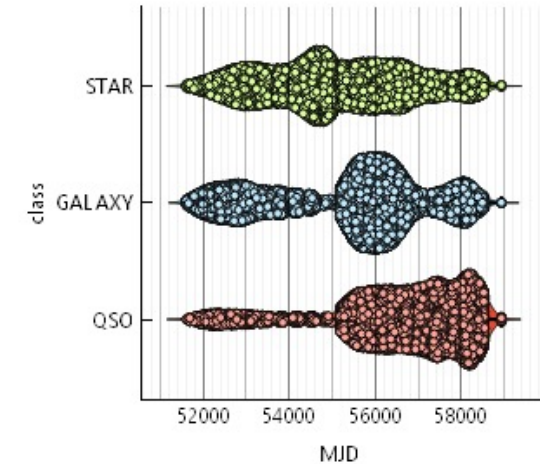
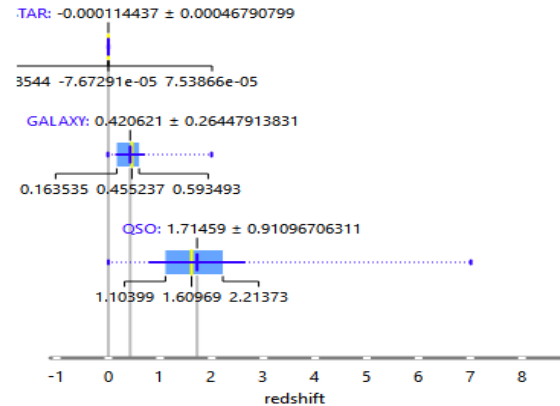
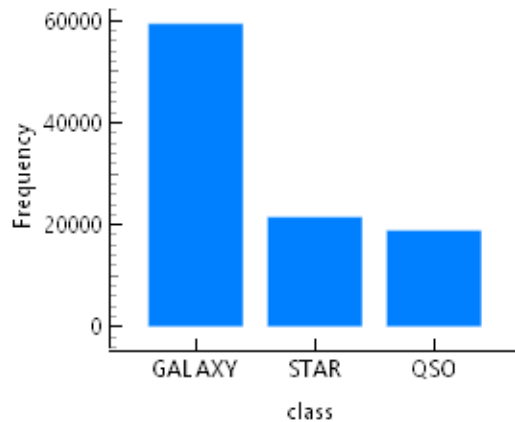
Alpha and delta denote sky coordinates and appear to be evenly distributed.

The photometric filters **u** and **g** display significant variation between classes, but include invalid entries (-9999) that require cleaning.

obj_ID serves as a unique identifier without predictive relevance and will be excluded from the analysis.

Feature distributions seem generally consistent across classes; however, quasars (QSOs) are underrepresented, suggesting potential class imbalance — this warrants further exploration during EDA.

Exploratory Data Analysis



- The dataset is predominantly composed of GALAXY objects.
- QSOs (quasars) are the least represented class, highlighting the need for class balancing techniques such as SMOTE.
- QSOs display notably higher redshift values compared to stars and galaxies.
- This observation supports established astronomical theory, as quasars are known to be some of the farthest and earliest objects in the universe.
- In contrast, the lower redshift values of stars and galaxies suggest they are much closer in distance.
- Observation dates are uniformly distributed over time across all three classes.
- However, QSOs and GALAXY instances show higher concentration during specific timeframes, possibly indicating periods of focused survey efforts.

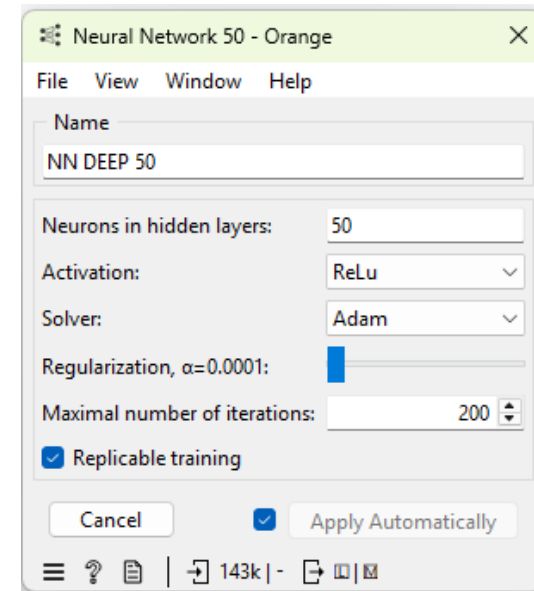
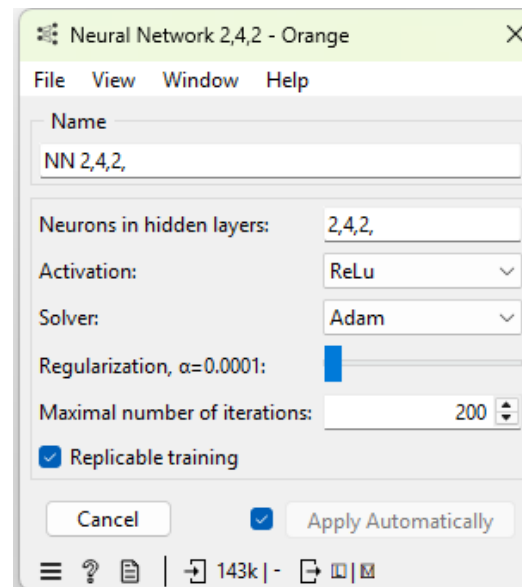
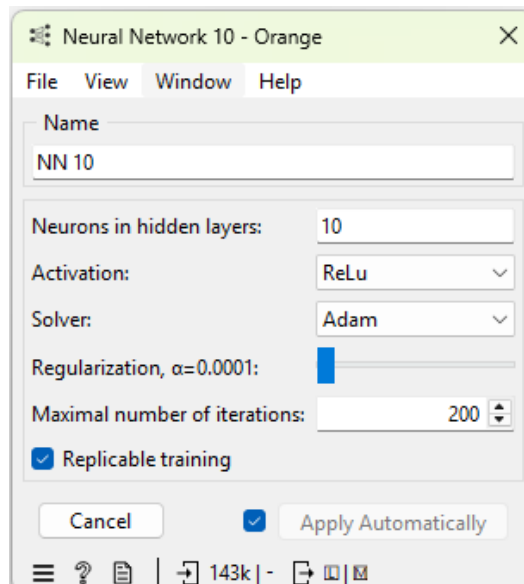
SMOTE and Undersampling

- Our dataset shows a pronounced class imbalance, with GALAXY objects overwhelmingly dominating the samples, while QSOs (quasars) and STARS are significantly underrepresented.
- Such imbalance can skew machine learning models—particularly neural networks—toward favoring the majority class, which in turn reduces recall and precision for minority classes like QSOs that are central to our research objectives.
- To address class imbalance, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied to generate synthetic examples for the underrepresented classes. This ensures the model is exposed to sufficient representative data, enhancing its ability to generalize and accurately identify minority classes.
- Additionally, **undersampling** was used to reduce the number of majority class (GALAXY) instances. This prevents the model from being overwhelmed by dominant class features, promoting balanced learning across all classes.

```
def python_script(in_data):
    2 import numpy as np
    3 from Orange.data import Domain, Table
    4 from imblearn.under_sampling import ClusterCentroids
    5 from collections import Counter
    6
    7 # Copy input data
    8 df = in_data.copy()
    9 X, y = df.X, df.Y
    10
    11 # Count class frequencies
    12 counts = Counter(int(val) for val in y)
    13 min_count = min(counts.values())
    14
    15 # Create strategy to downsample all classes to minority count
    16 strategy = {cls: min_count for cls in counts}
    17
    18 # Apply ClusterCentroids
    19 cc = ClusterCentroids(sampling_strategy=strategy)
    20 X_res, y_res = cc.fit_resample(X, y)
    21
    22 # Build domain and Orange table
    23 domain = Domain(df.domain.attributes, df.domain.class_vars)
    24 out_data = Table(domain, X_res, y_res)
    25
```

```
def python_script(in_data):
    1 import Orange
    2 import numpy as np
    3 from Orange.data import Domain, Table
    4 from imblearn.over_sampling import SMOTE
    5 from collections import Counter
    6
    7 # Ensure data is connected
    8 df = in_data.copy()
    9 X, y = df.X, df.Y
    10
    11 # Show class distribution before resampling
    12 print("Before SMOTE:", Counter(int(val) for val in y))
    13
    14 # Apply SMOTE to balance all classes
    15 smote = SMOTE()
    16 X_res, y_res = smote.fit_resample(X, y)
    17
    18 # Show class distribution after resampling
    19 print("After SMOTE:", Counter(int(val) for val in y_res))
    20
    21 # Create new Orange table
    22 domain = Domain(df.domain.attributes, df.domain.class_vars)
    23 out_data = Table(domain, X_res, y_res)
    24
```

Neural Network Architectures



1.NN 10 (Shallow Network)

Architecture: Single hidden layer with 10 neurons

Activation: ReLU

Solver: Adam

Max iterations: 200

Purpose: Acts as a baseline model — fast to train and useful for observing general trends in balanced data.

2.NN 2-4-2 (Deep + Narrow)

Architecture: Three hidden layers with 2, 4, and 2 neurons

Activation: ReLU

Solver: Adam

Purpose: Tests how narrow deep networks behave on a balanced dataset. Useful for identifying if deeper layers improve feature extraction without overfitting.

3.NN Deep 50 (Wide Network)

Architecture: Single hidden layer with 50 neurons

Activation: ReLU

Solver: Adam

Purpose: Enables the network to capture complex patterns across features — especially beneficial after SMOTEing when more minority class examples are present.

Model Evaluation

Three neural network architectures were evaluated on the balanced dataset (via SMOTE) using metrics such as AUC, Accuracy, F1-score, Precision, Recall, and MCC.

• **NN DEEP 50** outperformed all models with the highest scores across all metrics:

- **AUC = 0.997, F1 = 0.977, and CA (Accuracy) = 0.977**

• **NN 10** also achieved strong results with a slightly lower F1-score of **0.974**

• **NN 2,4,2** performed well but had the lowest precision and recall among the three

The **confusion matrices** show that:

- All models classified **GALAXY** and **STAR** with high accuracy
- **QSO (quasar)** classification showed improvement after SMOTE, with **NN DEEP 50** yielding the most balanced performance across all classes

Overall, **NN DEEP 50** is recommended for deployment due to its superior and consistent performance.

Model	AUC	CA	F1	Prec	Recall	MCC
NN 2,4,2,	0.994	0.972	0.971	0.972	0.972	0.957
NN 10	0.996	0.974	0.974	0.974	0.974	0.962
NN DEEP 50	0.997	0.977	0.977	0.977	0.977	0.966

		Predicted			Σ
		GALAXY	QSO	STAR	
Actual	GALAXY	45861	1135	614	47610
	QSO	1876	45691	5	47572
	STAR	7	5	47474	47486
	Σ	47744	46831	48093	142668

		Predicted			Σ
		GALAXY	QSO	STAR	
Actual	GALAXY	45781	1294	535	47610
	QSO	2227	45341	4	47572
	STAR	0	1	47485	47486
	Σ	48008	46636	48024	142668

		Predicted			Σ
		GALAXY	QSO	STAR	
Actual	GALAXY	46065	1005	540	47610
	QSO	1680	45887	5	47572
	STAR	16	0	47470	47486
	Σ	47761	46892	48015	142668

Key Findings

Best Performing Model:

The **NN DEEP 50** model—featuring a single hidden layer with 50 neurons—delivered the strongest results:

- Accuracy:** 97.7%
- F1-Score:** 0.977
- AUC:** 0.997

After applying SMOTE, the model demonstrated balanced performance across all classes, effectively addressing class imbalance.

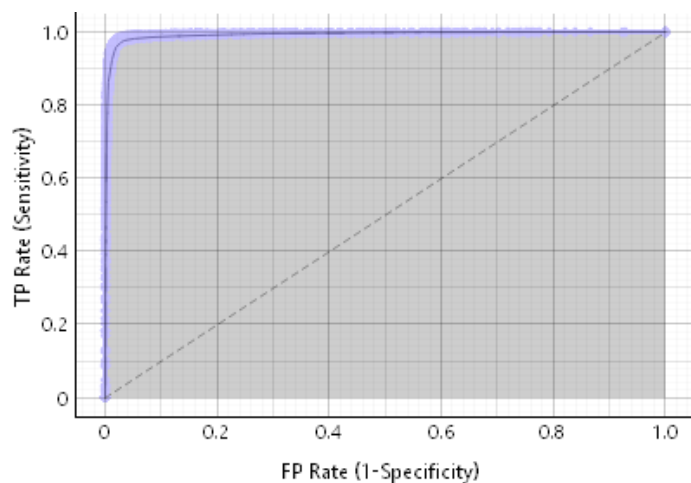
Challenges Encountered:

- The dataset suffered from significant class imbalance, with GALAXY instances vastly outnumbering QSOs (quasars).
- Initial neural network models were biased toward the majority class, resulting in poor performance on minority classes.
- Implementing **SMOTE** was essential to synthetically oversample underrepresented classes and enhance the model's generalization capability.

Misclassification Insights:

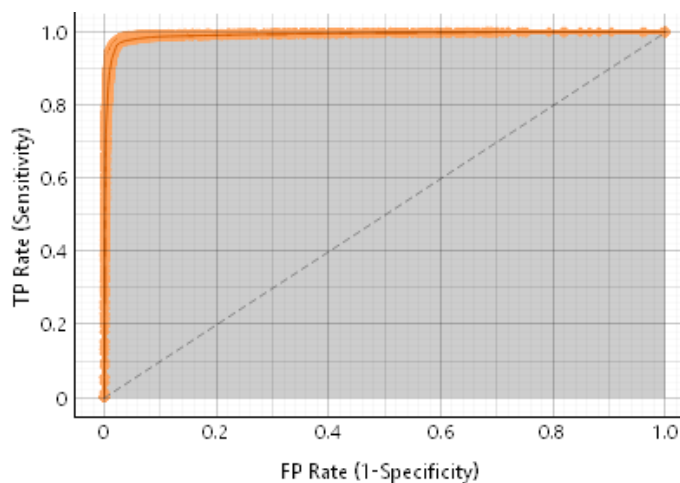
- Prior to class balancing, quasars were often misclassified as stars or galaxies due to limited training examples.
- After applying SMOTE, QSO classification improved markedly, though some misclassification with stars persisted, as reflected in the confusion matrices.
- This suggests that quasars and stars may exhibit overlapping spectral characteristics, making it challenging for the model to draw clear decision boundaries.

ROC Curves of 3 different Neural Networks Architectures



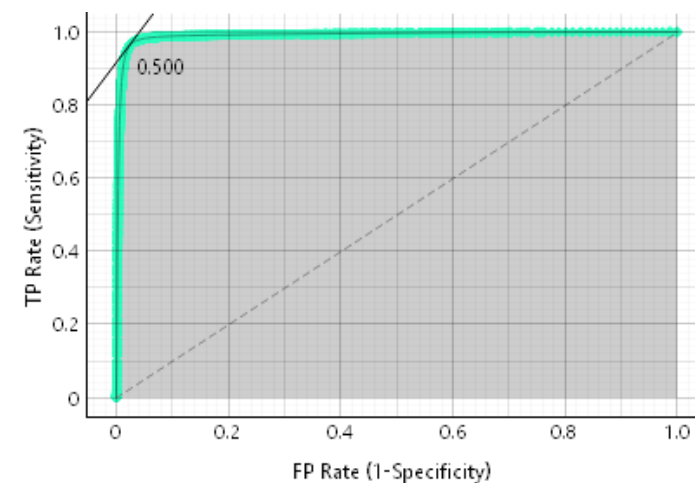
NN Deep-50

Exhibits the most ideal ROC curve, nearly hugging the top-left, indicating exceptional performance across all thresholds.



NN Simple 10

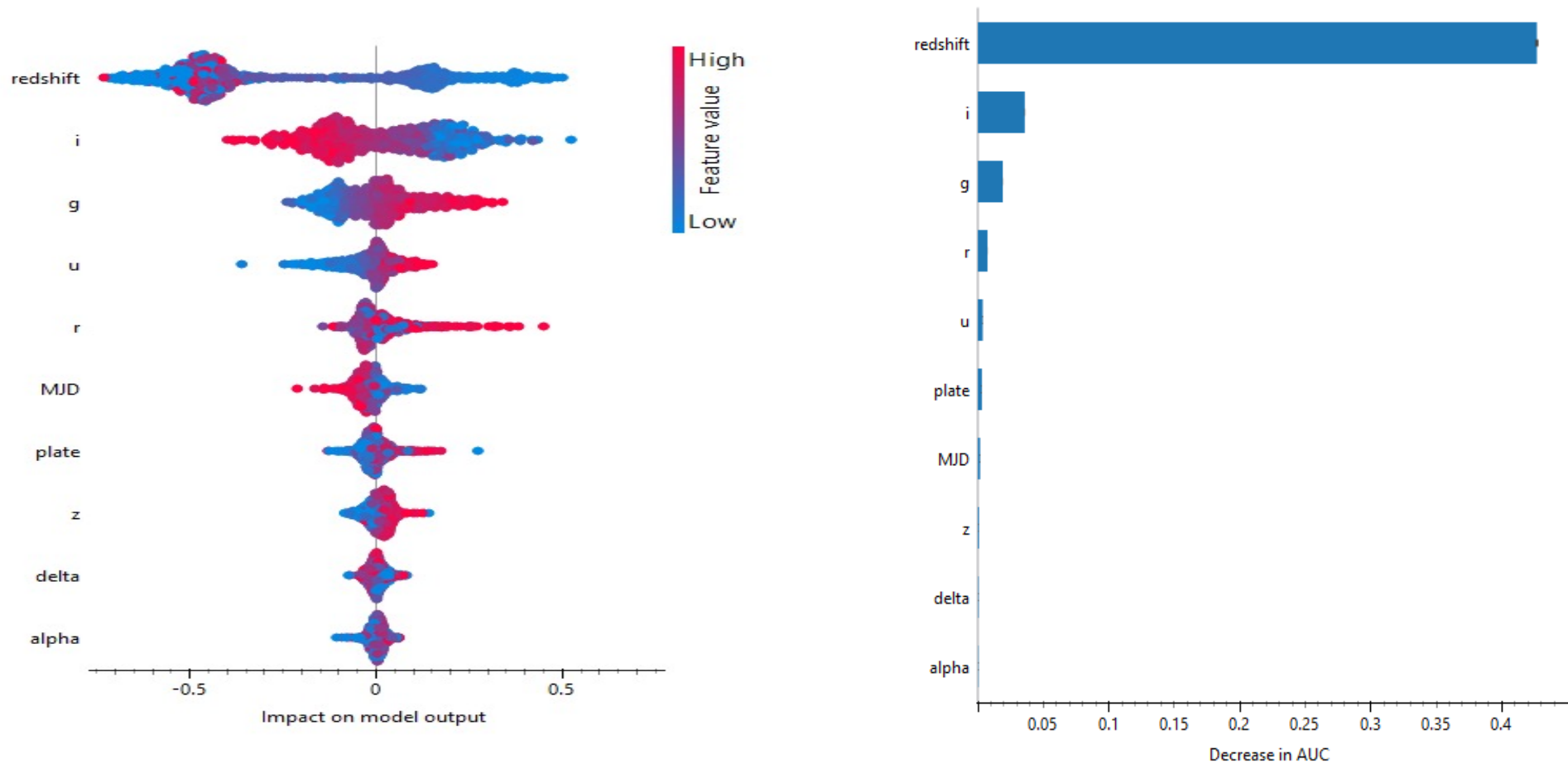
Also shows strong classification but with slightly **less curvature** compared to Deep-50, suggesting slightly reduced sensitivity.



NN DEEP AND NARROW WITH 3 HIDDEN LAYERS

Performs reasonably well, but the ROC curve is slightly less steep, with **AUC around 0.994**, indicating it's still effective but not as robust as Deep-50.

Feature Importance and Model Explainability



Feature Importance and Model Explainability

The visualizations highlight which features had the greatest impact on the neural network's ability to classify celestial objects.

- **SHAP Summary Plot**

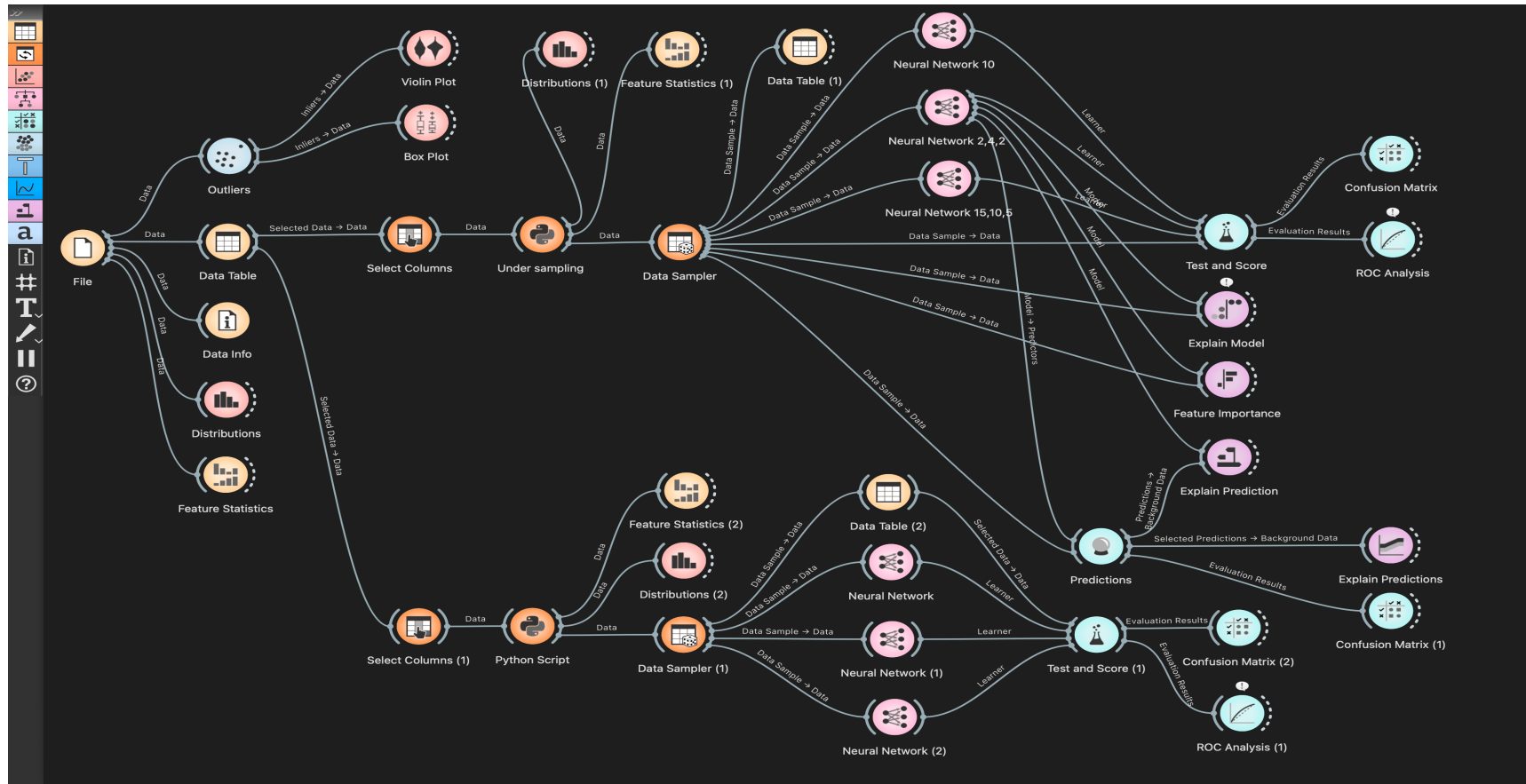
- **Redshift** is the most influential feature in determining the class of a celestial object.
- Higher redshift values strongly drive the prediction toward quasars (QSO).
- Photometric filters like i, g, u, and r also significantly contribute to classification decisions.

- **Feature Importance**

Removing or shuffling **redshift** caused the largest drop in model performance (AUC dropped the most), confirming it as the most critical feature.

- Other filters (i, g, u) show minor but notable importance.
- Positional features like alpha and delta had minimal impact on model accuracy.

Orange Workflow



Final Recommendations

Model Deployment Recommendations:

•Deploy the NN DEEP-50 Model:

•Among all tested architectures, the neural network with a single hidden layer of 50 neurons consistently outperformed others (**AUC = 0.997**, **F1 = 0.977**). Its strong performance makes it well-suited for accurate classification of celestial objects, particularly quasars.

•Integrate SMOTE into the Preprocessing Pipeline:

•Implementing SMOTE effectively addressed class imbalance, significantly enhancing recall and precision for rare classes like QSOs. It should remain a core component in all future preprocessing workflows.

•Prioritize Redshift and Photometric Features:

•Feature importance analysis highlights **redshift** and photometric bands (**i, g, u, r**) as key contributors to model predictions. Future models should focus on refining, engineering, or optimizing these features to further boost accuracy.

•Monitor QSO–Star Misclassification:

•Despite improvements, some quasars continue to be misclassified as stars, likely due to shared spectral traits. This warrants further investigation, possibly through the incorporation of more granular features or advanced ensemble learning techniques.